

Q1:- The tension T in the string of the yo-yo is

$$T = \frac{mgR}{2r^2 + R^2}$$

where ' m ' is the mass of yo-yo & g is acc due to gravity. Use differentials to estimate the change in the tension if R is increased from 3cm to 3.1cm & r is increased from 0.7cm to 0.8cm - Does the tension increase or decrease?

Sol

$$dT = \frac{\partial T}{\partial r} dr + \frac{\partial T}{\partial R} dR \rightarrow (a)$$

$$\frac{\partial T}{\partial r} = - \frac{mgR(4r)}{(2r^2 + R^2)^2}$$

$$\frac{\partial T}{\partial R} = - \frac{4mgRr}{(2r^2 + R^2)^2}$$

$$\frac{\partial T}{\partial R} = \frac{mg(2r^2 + R^2) - mgR(2R)}{(2r^2 + R^2)^2}$$

$$\frac{\partial T}{\partial R} = \frac{mg(2r^2 + R^2 - 2R^2)}{(2r^2 + R^2)^2}$$

$$\frac{\partial T}{\partial R} = \frac{mg(2r^2 - R^2)}{(2r^2 + R^2)^2}$$

Substituting $\frac{\partial T}{\partial r}$ & $\frac{\partial T}{\partial R}$ in (a)

$$dT = - \frac{4mgRr}{(2r^2 + R^2)^2} dr + \frac{mg(2r^2 - R^2)}{(2r^2 + R^2)^2} dR$$

$$dT \text{ or } \Delta T \approx - \frac{4mgRr}{(2r^2 + R^2)^2} \Delta r + \frac{mg(2r^2 - R^2)}{(2r^2 + R^2)^2} \Delta R \rightarrow (b)$$

$$\Delta r = 0.1, \Delta R = 0.1, r = 0.7, R = 3$$

Substituting values in (b) we get

$$\Delta T \text{ or } \Delta T \approx \frac{-4mg(3)(0.7)}{(2(0.7)^2 + (3)^2)^2} (0.1) + \frac{mg(2(0.7)^2 - (3)^2)}{(2(0.7)^2 + (3)^2)^2} (0.1)$$

$$\Delta T \approx -0.02648mg$$

Since ΔT is -ive, Tension decreases.

Q2:- If R is the total resistance of three resistors, connected in parallel, with resistance R_1, R_2, R_3 , then

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

If resistances are measured in ohms as $R_1 = 25\Omega, R_2 = 40\Omega$ & $R_3 = 50\Omega$, with a possible error of 0.5% in each case, estimate the max error in calculated value of R .

Sol

$$\Delta R = \frac{\partial R}{\partial R_1} \Delta R_1 + \frac{\partial R}{\partial R_2} \Delta R_2 + \frac{\partial R}{\partial R_3} \Delta R_3$$

$$\frac{\partial}{\partial R_1} \frac{1}{R} = \frac{\partial}{\partial R_1} \left(\frac{1}{R_1} \right) + \frac{\partial}{\partial R_2} \left(\frac{1}{R_2} \right) + \frac{\partial}{\partial R_3} \left(\frac{1}{R_3} \right)$$

$$-\frac{1}{R^2} \frac{\partial R}{\partial R_1} = -\frac{1}{R_1^2}$$

$$\frac{\partial R}{\partial R_1} = \frac{R^2}{R_1^2}$$

$$\frac{\partial R}{\partial R_2} = \frac{R^2}{R_2^2}$$

$$\frac{\partial R}{\partial R_3} = \frac{R^2}{R_3^2}$$

$$\Delta R_1 = dR_1 = 25 \times 0.005 = 0.125$$

$$\Delta R_2 = dR_2 = 40 \times 0.005 = 0.2$$

$$\Delta R_3 = dR_3 = 50 \times 0.005 = 0.25$$

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$\frac{1}{R} = \frac{1}{25} + \frac{1}{40} + \frac{1}{50}$$

$$\frac{1}{R} = \frac{17}{200}$$

$$R = 11.7647$$

$$\Delta R = \frac{\partial R}{\partial R_1} \cdot \Delta R_1 + \frac{\partial R}{\partial R_2} \cdot \Delta R_2 + \frac{\partial R}{\partial R_3} \cdot \Delta R_3$$

$$\Delta R = \frac{(11.7647)^2}{(25)^2} \times (0.125) + \frac{(11.7647)^2}{(40)^2} \times (0.2) + \frac{(11.7647)^2}{(50)^2} \times (0.25)$$

$$\Delta R \approx 0.059$$

Q3:- A rectangular beam that is supported at its two ends & is subjected to a uniform load sags by an amount

$$S = C \frac{Pl^4}{wh^3}$$

where P = load, l = length, h = height, w = width & C is a constant. Suppose $P \approx 100$, $l = 4$, $w \approx 0.1$ & $h \approx 0.2$. Will the sag of the beam be more sensitive to changes in the height of beam or changes in width of beam.

Sol $\Delta S \approx \frac{\partial S}{\partial P} \Delta P + \frac{\partial S}{\partial l} \Delta l + \frac{\partial S}{\partial w} \Delta w + \frac{\partial S}{\partial h} \Delta h$

$$= \dots$$

$$\frac{\partial S}{\partial P} = C \frac{l^4}{wh^3}$$

$$\frac{\partial S}{\partial l} = C \frac{4Pl^3}{wh^3}$$

$$\frac{\partial S}{\partial w} = -C \frac{Pl^4}{w^2h^3}$$

$$\frac{\partial S}{\partial h} = -C \frac{3Pl^4}{wh^4}$$

$$\Delta S = \frac{C l^4}{w h^3} \Delta P + \frac{C 4 P l^3}{w h^3} \Delta l - \frac{C P l^4}{w^2 h^3} \Delta w - \frac{C 3 P l^4}{w h^4} \Delta h$$

when $w \approx 0.1$ & $h \approx 0.2$

For sake of simplicity

$$\Delta S = \frac{C P l^4}{w h^3} \left[\frac{\Delta P}{P} + 4 \frac{\Delta l}{l} - \frac{\Delta w}{w} - 3 \frac{\Delta h}{h} \right]$$

$$\frac{\Delta w}{w} \approx 10 \Delta w \quad 3 \frac{\Delta h}{h} \approx 15 \Delta h$$

The sag is more sensitive to Δh .

Q4:- Find the linearization $L(x, y)$ of fcn $f(x, y) = e^x \cos y$ at $P_0(0, 0)$
~~and~~ Then find an upper bound for the magnitude $|E|$ of the error in the approximation of f over the rectangle $R: |x| \leq 0.1, |y| \leq 0.1$
 (use $e^x \leq 1.11$ & $|\cos y| \leq 1$)

Sol $f(0, 0) = 1$

$$f_x(x, y) = e^x \cos y \quad f_x(0, 0) = 1$$

$$f_y(x, y) = -e^x \sin y \quad f_y(0, 0) = 0$$

$$L(x, y) = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

$$L(x, y) = 1 + 1(x - 0) + 0(y - 0)$$

$$L(x, y) = x + 1$$

$$f_{xx} = e^x \cos y, \quad f_{yy} = -e^x \cos y, \quad f_{xy} = -e^x \sin y$$

$$|x| \leq 0.1$$

$$-0.1 \leq x \leq 0.1$$

$$|y| \leq 0.1$$

$$-0.1 \leq y \leq 0.1$$

$$|f_{xx}| = |e^x \cos y| \leq 1.11$$

$$|f_{yy}| = |-e^x \cos y| \leq 1.11$$

$$|f_{xy}| = |-e^x \sin y| \leq e^{0.1} \sin(0.1) \leq 0.12$$

we take $M=1.11$ as upper bound on second partials

$$|E| \leq \frac{1}{2} (1.11) (|x|+|y|)^2$$

$$\leq (0.555) (0.1 + 0.1)^2$$

$$0.0222$$

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Example 2. Let $f(x, y) = e^x \sin y$.

- a. Find the linearization $L(x, y)$ of f at $P_0 = P_0(\ln 2, \pi/2)$.

$$f_x = e^x \sin y, \Rightarrow f_x(\ln 2, \pi/2) = 2$$

$$f_y = e^x \cos y, \Rightarrow f_y(\ln 2, \pi/2) = 0$$

so that

$$\begin{aligned} L(x, y) &= f(\ln 2, \pi/2) + f_x(\ln 2, \pi/2)(x - \ln 2) + f_y(\ln 2, \pi/2)(y - \pi/2) \\ &= 2 + 2(x - \ln 2) \end{aligned}$$

- b. Find an upper bound for the magnitude $|E|$ of the error in the approximation $f(x, y) \approx L(x, y)$ over the rectangle

$$R: |x - \ln 2| \leq 0.1, |y - \pi/2| \leq 0.2. \quad (\text{use } |\cos y| \leq 1, |\sin y| \leq 1)$$

The error is bounded by the formula

$$|E| \leq \frac{M}{2} (|x - \ln 2| + |y - \pi/2|)^2$$

where M is an upper bound of all of the second order partials of f over the rectangle R . Now,

$$f_{xx} = e^x \sin y \Rightarrow |f_{xx}| = |e^x \sin y| \leq e^x \leq e^{\ln 2 + 0.1}, \quad (x, y) \in R$$

and since

$$f_{yy} = -e^x \sin y$$

$$f_{xy} = f_{yx} = e^x \cos y$$

we conclude that $M = e^{\ln 2 + 0.1}$. Thus

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$$\begin{aligned}
 |E| &\leq \frac{e^{\ln 2 + 0.1}}{2} (0.1 + 0.2)^2 \\
 &= \frac{e^{\ln 2 + 0.1}}{2} (0.09) \\
 &\leq \frac{2.4}{2} (0.09) = 0.108
 \end{aligned}$$

- c. Use the linearization of $f(x, y)$ from part (a) to estimate $f(0.75, 1.5)$.
We have

$$\begin{aligned}
 f(0.75, 1.5) &\approx L(0.75, 1.5) \\
 &= 2 + 2(0.75 - \ln 2) \\
 &\approx 2 + 2(0.75 - 0.693) \\
 &\approx 2.114
 \end{aligned}$$

According to MMA $f(0.75, 1.5) \approx 2.1117 \pm 0.0001$.

2. Consider the surfaces $S_1: z = x^2 - y^2$ and $S_2: y^2 + z^2 = 10$

(a) (10 points) Find an equation of the tangent line to the curve of intersection of S_1 and S_2 at the point $(2, 1, 3)$.

(b) (5 points) Find the acute angle between the planes which are tangent to the surfaces S_1 and S_2 at the point $(2, 1, 3)$.

$$\begin{aligned} \text{(a)} \quad S_1: \text{let } f(x, y, z) &= z - x^2 + y^2 & S_2: \text{let } g(x, y, z) &= y^2 + z^2 \\ \nabla f(x, y, z) &= \langle -2x, 2y, 1 \rangle & \nabla g(x, y, z) &= \langle 0, 2y, 2z \rangle \\ \nabla f(2, 1, 3) &= \langle -4, 2, 1 \rangle & \nabla g(2, 1, 3) &= \langle 0, 2, 6 \rangle \end{aligned}$$

$$\begin{aligned} \text{Let } \vec{n} &= \nabla f(2, 1, 3) \times \nabla g(2, 1, 3) \\ &= \langle -4, 2, 1 \rangle \times \langle 0, 2, 6 \rangle \\ &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -4 & 2 & 1 \\ 0 & 2 & 6 \end{vmatrix} = (12-2)\vec{i} - (-24-0)\vec{j} + (-8-0)\vec{k} \\ &= 10\vec{i} + 24\vec{j} - 8\vec{k} \end{aligned}$$

So the tangent line $\vec{r}(t)$ is

$$\begin{aligned} \vec{r}(t) &= \langle 2, 1, 3 \rangle + t \vec{n} \\ &= \langle 2, 1, 3 \rangle + t \langle 10, 24, -8 \rangle \end{aligned}$$

$$\text{or } x(t) = 2 + 10t, \quad y(t) = 1 + 24t, \quad z(t) = 3 - 8t$$

$$\begin{aligned} \text{(b)} \quad \cos \theta &= \frac{\langle -4, 2, 1 \rangle \cdot \langle 0, 2, 6 \rangle}{\| \langle -4, 2, 1 \rangle \| \| \langle 0, 2, 6 \rangle \|} \\ &= \frac{-4 \cdot 0 + 2 \cdot 2 + 1 \cdot 6}{21 \cdot 40} = \frac{1}{84} \end{aligned}$$

$$\Rightarrow \theta = \cos^{-1}\left(\frac{1}{84}\right)$$